Project Design Phase-II
Data Flow Diagram & User Stories

Date	31 January 2026
Team ID	
Project Name	
Maximum Marks	4 Marks

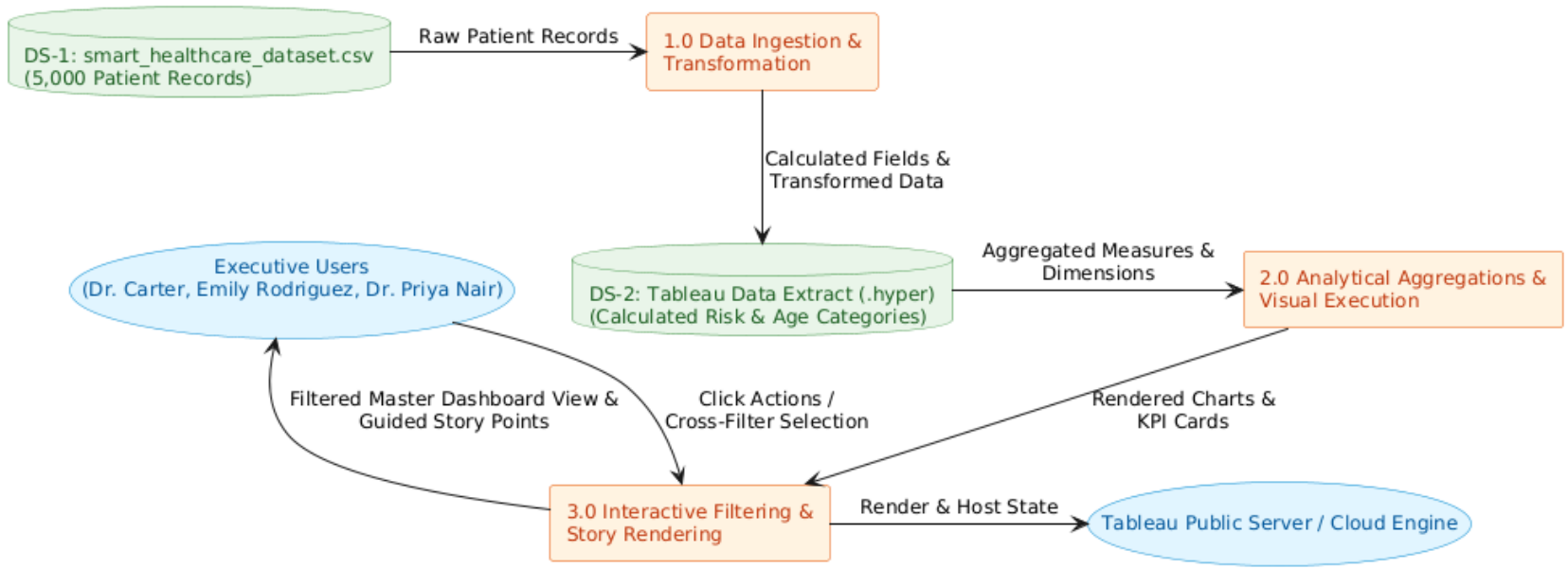
Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	**External Entity**	• **Healthcare Stakeholders / Executive Users**: Dr. Michael Carter (CMO), Emily Rodriguez (Wellness Director), Dr. Priya Nair (Head of Population Health). • **Tableau Public Server / Cloud Engine**: Hosting platform delivering web and story rendering
Rectangle with numbered header	**Process**	• **Process 1.0 (Data Ingestion & Transformation)**: Ingests raw CSV data and calculates row-level metrics (Age Category classification, composite risk scoring). • **Process 2.0 (Analytical Aggregations & Visual Execution)**: Processes measures and dimensions across 9 dedicated worksheets (scatter plots, heatmaps, bar charts, pie charts). • **Process 3.0 (Interactive Filtering & Story Rendering)**: Handles user interaction (cross-filtering, parameter selection) and presents the 9-point Tableau Story narrative.

Rectangle (solid fill, no number)	**Data Store**	• **DS-1: smart_healthcare_dataset.csv**: Raw flat file containing 5,000 anonymized patient records (vitals, symptoms, habits, demographics). • **DS-2: Tableau Data Extract (.hyper)**: High-performance columnar dataset containing custom calculated fields (Age Category, Calculated Risk Score, binary disease aggregations).
Labelled arrow	**Data Flow**	• **smart_healthcare_dataset.csv (DS-1) → *(Raw Patient Records)* → Process 1.0** • **Process 1.0 → *(Calculated Fields & Transformed Data)* → Tableau Data Extract (DS-2)** • **Tableau Data Extract (DS-2) → *(Aggregated Measures & Dimensions)* → Process 2.0** • **Process 2.0 → *(Rendered Charts & KPI Cards)* → Process 3.0** • **Executive User → *(Click Actions / Cross-Filter Selection)* → Process 3.0** • **Process 3.0 → *(Filtered Master Dashboard View & Guided Story Points)* → Executive User** • **Process 3.0 → *(Render & Host State)* → Tableau Cloud**



User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Chief Medical Officer (Dr. Michael Carter)	Clinical Risk Evaluation	USN-1	As CMO, I want to evaluate Blood Pressure vs. Cholesterol on a scatter plot overlayed with composite risk scores, so that I can identify high-risk cardiovascular patients.	Scatter plot renders with reference lines and color gradient representing risk scores (0–5).	High	Sprint-1
Chief Medical Officer (Dr. Michael Carter)	Disease Case Tracking	USN-2	As CMO, I want a bar chart quantifying chronic disease counts (Diabetes, Heart Disease, Stroke), so that I can measure chronic condition prevalence across weight classes.	Disease counts clearly display numbers (e.g., 660 Diabetes, 550 Heart Disease) segmented by BMI category.	High	Sprint-1
Director of Wellness Programs (Emily Rodriguez)	Behavioural Risk Matrix	USN-3	As Wellness Director, I want a multi-factor lifestyle heatmap cross-tabulating smoking, exercise, and alcohol usage, so that I can spot high-risk lifestyle combinations.	Matrix calculates average risk scores across 12 distinct habit combinations.	High	Sprint-1
Director of Wellness Programs (Emily Rodriguez)	Physical Activity Impact	USN-4	As Wellness Director, I want to analyse average BMI across physical activity levels (Sedentary, Moderate, Active), so that I can target lifestyle intervention programs.	Grouped bar chart displays average BMI values per age category across exercise levels.	Medium	Sprint-2
Head of Population Health (Dr. Priya Nair)	Demographic Stratification	USN-5	As Head of Population Health, I want an age and gender distribution breakdown across 6 life stages (Children to Elder), so that I can allocate healthcare resources accurately.	Population breakdown sorted chronologically by age category and segmented by gender.	High	Sprint-1
Head of Population Health (Dr. Priya Nair)	Symptom Prevalence Tracking	USN-06	As Head of Population Health, I want to track primary symptom counts (Chest Pain, Dizziness, Fatigue), so that I can anticipate clinical staffing demands.	Horizontal/Vertical bar chart displays integer symptom counts per age cohort without decimal places.	Medium	Sprint-2
Executive User (All Personas)	Interactive Drill-Down	USN-07	As an executive user, I want dynamic 1-click cross-filtering across the Master Dashboard, so that selecting a segment updates all 9 visualizations simultaneously without query lag.	Dashboard filter action updates total patient count and all sub-charts in under 2 seconds.	High	Sprint-1

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Executive User (All Personas)	Executive Storytelling	USN-08	As an executive user, I want a guided 9-point Tableau Story narrative, so that I can present strategic findings sequentially to clinical board members.	Story tab contains 9 navigation points with concise executive titles and captions.	High	Sprint-2